
Recognizing the predictive boundary of historically restricted physical knowledge

Chenxi He*

Cavendish Laboratory
University of Cambridge
Cambridge CB3 0HE, United Kingdom

Jingqiu Chen

College of Physics and Optoelectronic Engineering
Shenzhen University
Shenzhen 518060, China

Abstract

Scientific models have finite domains, normally established only in hindsight, once a successor theory explains the failure. Machine-learning systems for scientific data are optimized to interpolate, to predict forward, or to recover a governing expression; none takes a declared body of physical knowledge as the object under test. Here we show that the predictive boundary of a body of physical knowledge can be tested from that knowledge alone. We train a model only on physical relations recorded before a chosen date, the knowledge horizon, give it no later formula and no example of a departure, and require it to rank relations that the earlier knowledge cannot account for above relations that it can. With the horizon at 1899, a neural encoder trained on pre-1900 laws alone, and given at inference only a point cloud with no formula, units or labels, ranks 55 twentieth-century law families above nine held-out pre-1900 groups with an area under the ROC curve (AUROC) of 0.925, against 0.649 for the same network before optimization. A second test, fixed in advance, fits four simple forms where a pre-1900 form fits the data and measures how far the remaining data depart from the extrapolation; it reaches 0.888 against 11 held-out pre-1900 families with no learned representation. When a named pre-1900 law, the incumbent, is tested directly against historical measurements, the discrepancy is tied to that law rather than to the data: on five such measurements it collapses under the successful successor prediction in four cases and under none of the five wrong-shape or five wrong-mechanism alternatives. The same construction applies to any period with a dated corpus, an admitted prediction and a calibration set.

1 Introduction

Physics advances by retiring descriptions that have stopped working. The Rayleigh–Jeans law, the Dulong–Petit law and the classical account of electrical resistance were each accurate across the range in which they were built and each failed outside it, and in each case the failure was explained only by a successor theory. The domain over which a model can be trusted is therefore something the field has established in retrospect.

*Correspondence: ch2067@cam.ac.uk

Machine-learning systems for scientific data are built for other targets. They are optimized to interpolate within a sampled range, to predict forward, or to recover a compact governing expression from observations [1, 2]. Conformal prediction calibrates residuals [3], change-point methods locate distributional shifts [4], and Bayesian model discrepancy compares observations with a simulator [5]. Symbolic regression searches for a replacement expression once a discrepancy has already been found [6, 7].

None of these operations takes a dated body of physical knowledge as the object under test. A residual is large relative to some fitted model, a change point is a shift in some distribution, and a recovered expression is a fit to some data; in each case the knowledge being audited is implicit in the setup. Nor does an unusual curve shape identify the departure, because a departure from a named law may keep an ordinary functional form while a familiar mechanism may produce an unusual one.

We call the date that bounds a body of knowledge its knowledge horizon: only relations recorded before that date are admitted, and the pre-horizon law tested against a given dataset is the incumbent. Here we make the knowledge horizon an explicit input and test it directly with EPOCH (Epistemic Probe of Classical Horizons). Every test states in advance which laws or functional forms it may fit, which parameters it fits, how the measurement errors and any censoring are modelled, which data it fits, every data-dependent choice it makes, and the statistic and error rate it reports (Methods). It returns one of four verdicts: compatible over the measured range; localized incompatibility, with an interval for the boundary; whole-range incompatibility, with no identifiable boundary; or abstention. Each test reports three kinds of evidence separately, never as a single averaged score. The learned component is a neural encoder trained only on pre-horizon relations, with no example of a departure, which scores a new curve by its distance from them. The four-form component fits one of four simple forms (power, exponential, linear or constant) to the part of a curve where an admitted pre-horizon form fits well, the interior, and measures how far the remaining points, the frontier, depart from the extrapolation. The theory-conditioned component compares the measurements with the incumbent prediction and calibrates the residuals against a null distribution simulated from that prediction on the same grid with the same errors.

We set the horizon at 1899, the eve of the replacement of the classical model set by quantum and relativistic physics and a date at which the classical corpus can be reconstructed from dated primary sources. Every relation admitted to training is traced to a dated primary source and assigned to a law family, the set of relations that share one law; training, validation and test data are divided by family, and no law appears on both sides. Training uses no later formula and no synthetic breakdown example, and the training and scoring protocol was frozen, that is fixed and time-stamped, before any score on later physics was computed. We find that an encoder built this way ranks twentieth-century law families above pre-1900 families withheld from training, and that the four-form component reaches the same ranking without any trained network. On five historical datasets the discrepancy with the pre-1900 prediction collapses under the successor prediction in four of the five and under no wrong-shape or wrong-mechanism alternative. On real measured series the two highest learned scores belong to instrument records, a pattern consistent with the observation process setting the largest score differences.

2 Results

2.1 Declaring the object of a boundary test

EPOCH returns one of four verdicts for a set of measurements and a declared set of admissible laws: compatible over the measured range; localized incompatibility, with an interval for the boundary; whole-range incompatibility, with no identifiable boundary; or abstention, when the evidence is insufficient (Fig. 1). A fixed routing rule decides which component produces the verdict. The four-form component is used when the measurements contain a contiguous stretch, the interior, over which a pre-1900 law fits well. The theory-conditioned component is used otherwise, when an incumbent prediction and a model of the measurement errors are available. When neither condition is met the verdict is abstention.

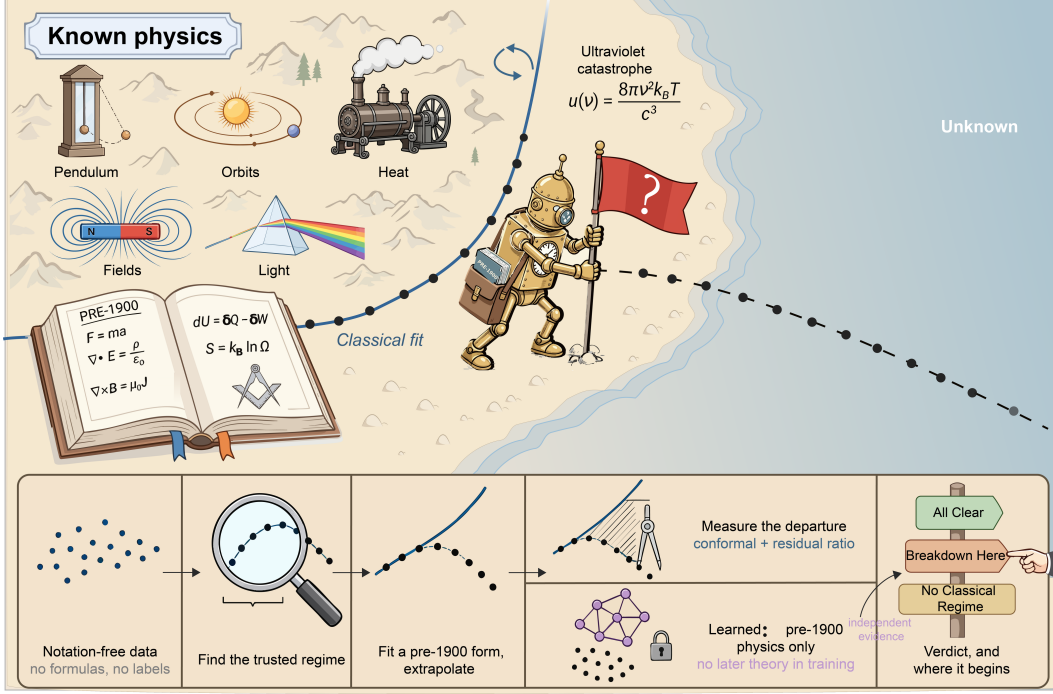


Figure 1. Testing a declared knowledge horizon. Top, an admitted classical form fitted inside the sampled regime (blue) tracks the measurements until they leave it; beyond that point the extrapolation and the data separate, and the question is whether the separation can be measured and located. Bottom, the test as applied here. A point cloud, a set of (x, y) pairs with no formula, unit or label, enters at the left. A fixed rule identifies the interior, the stretch over which an admitted pre-1900 form fits, and fits that form there; the extrapolation is compared with the frontier by two residual statistics (Methods); and an encoder trained on pre-1900 relations alone supplies an independent distance score. The signpost carries the verdict and, where an interior is present, the location at which compatibility is lost; a fourth output, abstention, applies when neither component can be run.

Each component is calibrated against its own reference distribution, the four-form and learned components against their scores on pre-1900 data and the theory-conditioned component against a null distribution simulated from the incumbent prediction, and the three are reported separately. The four-form component receives a point cloud, a set of (x, y) pairs with no formula, units or labels. It fits the best of four generic forms (power, exponential, linear and constant) on the interior and extrapolates the fit to the frontier, the remainder of the data. Two statistics measure the failure of the extrapolation. The split-conformal nonconformity is the mean absolute frontier residual divided by the 90th percentile of the interior absolute residuals. The extrapolation-residual ratio is the median absolute log-residual at the frontier divided by its interior value. Both denominators carry a one-percent relative floor (Methods). The theory-conditioned component computes the residuals of the measurements from the incumbent prediction, scaled by the declared measurement errors, and compares their size with the null distribution obtained by simulating data from that prediction on the same grid with the same errors. The learned component is a neural encoder trained only on point clouds from relations recorded before the horizon, with no example of a departure; its score is the distance of a new point cloud’s embedding from the embeddings of the training relations. A conclusion about a named physical theory requires that the admissible laws and the observation model encode that theory’s prediction for the measured quantity (Methods). Table 1 collects the principal quantitative results reported below, and Methods opens with the terms used.

Table 1. Principal quantitative results. Native controls are the nine pre-1900 groups as originally generated; generator-matched controls are the same groups regenerated with the sampling and noise process of the later families; the generator-gated set draws later formulas from the internal archive by the same generator as the pre-1900 corpus; metrological groups are calibration artefacts or instrument transfer curves. The 38 and 55 later families are the curated, citation-anchored set.

Analysis	Result
Large encoder, native controls, 38 families from 1901–1950 vs nine pre-1900 groups	AUROC 0.921 [0.833, 0.994]; before optimization 0.589
Large encoder, native controls, 55 families from 1901–1996 vs nine pre-1900 groups	AUROC 0.925 [0.851, 0.982]; before optimization 0.649
Large encoder, generator-matched controls (registered after the primary scores were opened), 55 later vs nine pre-1900 groups	AUROC 0.857 [0.741, 0.945]; before optimization 0.545
Large encoder, native generator-gated set of 1901–1950 formulas vs nine pre-1900 groups	AUROC 0.536 [0.424, 0.643]; before optimization 0.594 [0.436, 0.756]
Four-form component, frozen in advance, 55 later vs 11 pre-1900 families	Four-form AUROC 0.888 [0.800, 0.959]; compact encoder 0.669 [0.534, 0.798]
Compact encoder trained on pre-1900 laws vs generic-function control encoder, 12 breakdown and 12 classical families	Family AUROC 0.903 [0.729, 1.000] vs 0.375 [0.167, 0.618]
Frozen evaluation at the 0.5 screening threshold, 200 formula records and 30 synthetic breakdowns	AUROC 0.88
Four-form component at the calibrated threshold, 12 breakdown and 12 classical families	True-positive rate 0.271 [0.073, 0.500]; 0/96 false positives at the 5%-targeted threshold
Advancing the horizon on identical data grids	4/5 old-horizon discrepancies clear under the successful successor prediction
Wrong-shape and wrong-mechanism controls	5/5 wrong-shape and 5/5 wrong-mechanism predictions remain rejected
Frozen components on sixteen measured series, 5 post-1900 vs 9 classical or metrological groups	Learned AUROC 0.62 [0.29, 0.89]; four-form AUROC 0.78 [0.38, 1.00]; both intervals include 0.5
Boundary localization on 900 held-out simulations	90% interval coverage 91.3% [89.3%, 93.0%]
Simulated instrument effects at maximum severity	False-positive rate 0.40 for saturation; 0.63 for censoring; ≤ 0.11 for all other tested effects

2.2 Recognizing later physics from pre-1900 relations alone

To ask whether later physics can be ranked above pre-1900 physics from the shape of a relation alone, we trained a 20.9-million-parameter encoder (the large encoder) on 880,896 point clouds derived from 3,441 audited formula reductions of pre-1900 laws, each a dimensionless one-dimensional relation. During training, a second network that reads each formula’s expression tree produces a target embedding, and the point-cloud encoder, which is invariant to the order of the points, is trained to match it. At inference the encoder receives a normalized point cloud and nothing else, with no formula, graph, units, date, source or anomaly label. Every scoring choice was frozen before any score on later physics was computed (Methods).

Against nine held-out pre-1900 groups, each a set of law families sharing one mechanism, the learned score separates 38 families dated 1901–1950, each traced to a primary citation, with AUROC 0.921 [0.833, 0.994], 17 post-1950 families with 0.935 [0.823, 1.000] and their 55-family union with 0.925 [0.851, 0.982] (Fig. 2a; Fig. 3a). The same network with its initial random weights and no training steps (the model before optimization) gives 0.589, 0.784 and 0.649 (Fig. 3c). All four trained score components separate the 55-family union on their own, with AUROCs from 0.800 to 0.925 (Fig. 3d).

However, sampling alone could in principle produce that separation, because the later families were generated by a different process from the internal pre-1900 corpus. We therefore regenerated all nine internal groups, the generator-matched controls, with the point count, x-jitter, repeat count, noise range and heteroscedastic profile of the benchmark of 55 later

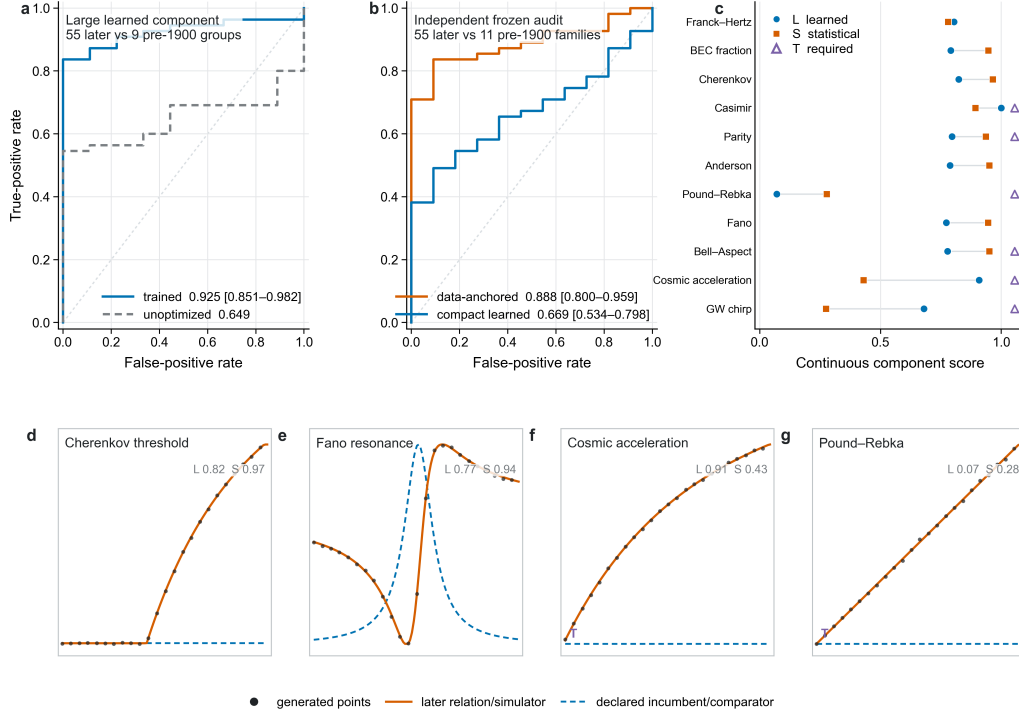


Figure 2. Temporal transfer and component scores across later physics. **a**, ROC curves for the large encoder trained on pre-1900 point clouds (blue) and the same network before optimization (grey), comparing 55 later law families with nine pre-1900 groups. **b**, Independent evaluation of the four-form component (orange) and compact encoder (blue) against 11 cited pre-1900 families. **c**, Learned scores (L, circles) and four-form percentiles against size-matched calibration (S, squares) for 11 cases assembled after the primary scores were computed (Supplementary Note 17). Triangles mark cases routed to the theory-conditioned component (T). **d–g**, Point clouds, later relations and incumbent predictions for Cherenkov emission [8], Fano resonance [9], cosmic acceleration [10] and Pound–Rebka redshift [11]. Checkpoints and calibration objects remain fixed throughout.

law families described in the next subsection; this leaves AUROC at 0.885 [0.778, 0.965] for the 38 near-future families and 0.857 [0.741, 0.945] for all 55, against 0.516 and 0.545 before optimization (Fig. 3a,b). Within the curated benchmark, pre-1900 training therefore contributes beyond the difference in generating process. A separate native generator-gated set draws 1901–1950 formulas from the internal archive by the same generator as the pre-1900 corpus (Supplementary Note 16). On that set the trained model gives 0.536 [0.424, 0.643] against 0.594 [0.436, 0.756] before optimization, so the curated and native results are not interchangeable.

For the compact encoder, evaluated on a separate 192-dataset benchmark of 12 breakdown and 12 classical families, the gain is specific to the historical corpus. A breakdown family is a set of synthetic curves built by imposing one kind of departure, such as saturation, a step or a kink, on a classical form; such curves are used only for evaluation, never for training or for setting thresholds. A source-audited reconstruction supplies 268 point clouds from 51 cited pre-1900 law families, split by family into 205/17/46 training, validation and test records. A compact encoder trained on the 205 classical records (the compact encoder throughout, trained with no unit information) reaches dataset-level AUROC 0.854 by k-nearest-neighbour distance and 0.895 by shrinkage Mahalanobis distance, against 0.329 and 0.288 for matched random features, and family-level AUROC 0.903 [0.729, 1.000]. A generic-function control encoder trained instead on generic smooth expressions, with identical architecture, seed, schedule, record count, family-size profile and functional-signature histogram, reaches 0.375 [0.167, 0.618]; the paired difference is 0.528 [0.264, 0.764]. Across three paired seeds the two ranges are 0.889–0.910 and 0.292–0.403. At the frozen family threshold the encoder detects

Classical-only representation learning at the 1899 knowledge horizon

One frozen pre-1900 model; later families are evaluation only; no anomaly templates, threshold selection, units, dates or source labels reach the query encoder.

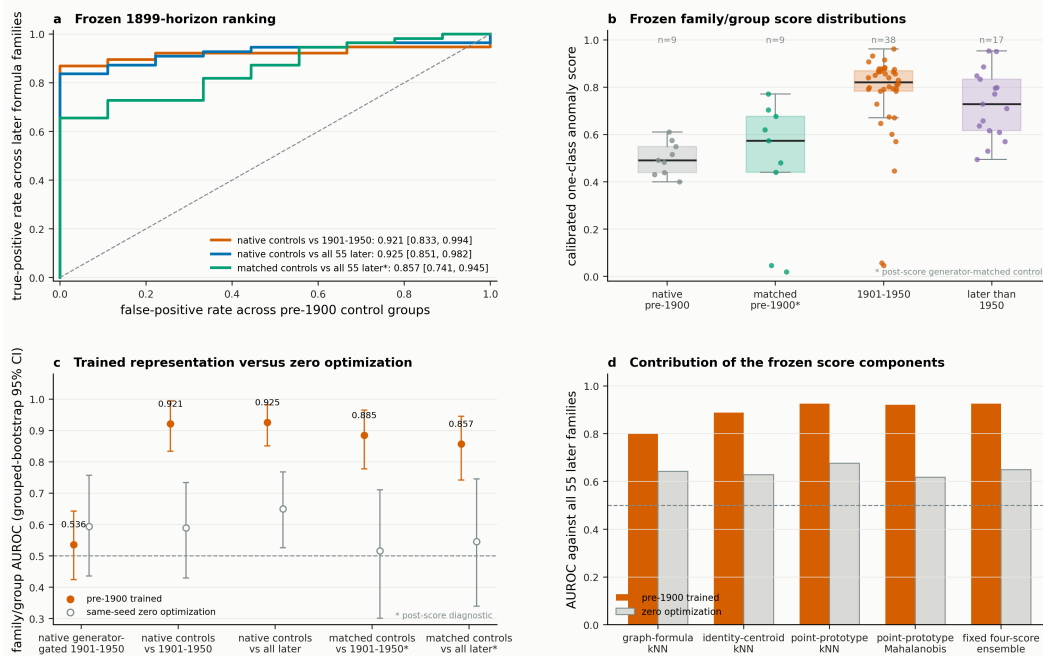


Figure 3. Large encoder at the 1899 horizon. **a**, The large encoder separates 38 families dated 1901–1950 and all 55 later families from nine pre-1900 groups. All-later AUROC is 0.857 [0.741, 0.945] against generator-matched controls. **b**, Learned score distributions for native and generator-matched pre-1900 controls and the two later-era strata. **c**, AUROCs for the trained network and the same network before optimization. The native generator-gated comparison is shown separately. **d**, Each of the four distance components and their fixed equal-weight mean improves over the network before optimization on the 55-family comparison. AUROCs and intervals are computed over law families or frozen relation groups.

2/12 breakdown families while clearing all classical controls, and this count is not stable across the three seeds (2/12, 8/12 and 1/12; Supplementary Note 8). A deformation control that retains each record’s physical units gives source-family-balanced AUROC 0.684 with dimensional labels and 0.922 without them, so the zero-unit ranking, obtained with no unit information supplied, is not carried by units.

2.3 Explicit extrapolation as independent evidence

Explicit extrapolation ranks later families above held-out pre-1900 families without a learned representation. The 55-family benchmark contains 55 named physical-law families first valid from 1901 to 1996, each traced to a primary citation. Eight independently sampled and noisy 120-point records of each give 440 formula-generated point clouds (Fig. 2b); eleven cited pre-1900 families excluded from encoder training are the controls. Both components are used frozen, and all quantities are reduced to one median per law family (Methods).

The four-form component ranks later families above the earlier holdouts with AUROC 0.888 [0.800, 0.959], and the compact encoder with 0.669 [0.534, 0.798]. An evaluation with all components frozen beforehand, over 200 formula records and 30 synthetic records of five breakdown families, gives 0.88, with recall 0.90 and a false-positive rate of 0.24 at the 0.5 screening threshold, the median of the calibration scores, which selects cases for further testing.

The ranking varies with functional form. Learned and four-form AUROCs are 0.629/0.795 for 12 later affine, power or exponential laws, 0.600/0.982 for five threshold or piecewise families, and 0.909/0.864 for six nonmonotone or oscillatory families, so the two scores respond differently to smooth familiar forms, to thresholds and to nonmonotone curves.

When the frozen components are applied to 11 further twentieth-century cases (Fig. 2c–g), the two components diverge again: Casimir plates [12] give 1.000/0.892, Cherenkov emission [8] 0.823/0.965, cosmic acceleration [10, 13] 0.908/0.429 and Pound–Rebka redshift [11] 0.071/0.277. Scalar or sparse-level claims, including the Lamb shift [14] and the electron anomalous magnetic moment [15], route to the theory-conditioned component instead, and the primary AUROC is unchanged (Supplementary Note 17). The calibrated detection rate of the four-form component at a fixed false-positive rate is reported below (Calibration, thresholds and the observation process).

2.4 Historical measurements at the 1899 horizon

Four historical datasets illustrate two of the four verdicts, whole-range incompatibility and localized incompatibility (Fig. 4a–d); a fifth, Millikan’s photoelectric measurement, is treated separately below. The four-form component’s initial screen gives percentiles (the fraction of size-matched classical calibration simulations scoring lower) of 0.95 for FIRAS, 0.95 for copper specific heat, 0.66 for Bertozzi and 0.66 for Onnes; whether an interior is present then determines which of the two outputs applies. Supplementary Note 4 gives finite-sample tail probabilities and the multiplicity calculation.

In the 43-point COBE-FIRAS cosmic-microwave-background monopole spectrum [16], a Rayleigh–Jeans law [17, 18] proportional to ν^2 over-predicts measured intensity above the peak by 475-fold to 3,000-fold across fitting windows; the 20% window gives a bootstrap interval of [1,500, 2,400]. At $T = 2.725$ K, FIRAS samples $x = h\nu/kT$ from 1.2 to 11.3, outside the Rayleigh–Jeans regime $x \ll 1$, so the output is whole-range incompatibility with the declared classical prediction.

Bertozzi’s electron measurements [19] report kinetic energy against $\beta^2 = v^2/c^2$. The classical law $\frac{1}{2}mv^2$ crosses $\beta^2 = 1$ at 0.26 MeV and reaches $\beta^2 = 58.7$ at 15 MeV, equivalent to $7.7c$, whereas the measured β^2 values approach 1. The lowest point is already $0.87c$, placing the complete series outside a non-relativistic interior, and the output is whole-range incompatibility with the Newtonian energy–velocity prediction.

The Dulong–Petit law [20] predicts temperature-independent molar heat capacity, whereas the NIST cryogenic reference curves for six solids fall at low temperature. Five materials contain a high-temperature plateau, the interior, and support localized incompatibility relative to the constant model; beryllium gives whole-range incompatibility. Interior R^2 ranges from 0.965 to 0.998 and extrapolation-residual ratios from 127 to 1,490, and for oxygen-free high-conductivity copper the classical extrapolation over-predicts the 4 K reference value by a factor of 3.7×10^3 . The reference-polynomial curves share one low-temperature mechanism and are grouped accordingly.

Onnes’s mercury resistance values [21], digitized from $R(T)$ and anchored by the reported resistance below $10^{-5} \Omega$, show a smooth normal-state trend followed by a discontinuous drop. Relative to the declared smooth and residual-resistance models [22], EPOCH localizes a candidate boundary at 4.20 K, with figure-reading uncertainty and the censored lower values included in the observation model.

In Millikan’s 1916 photoelectric measurement the shape evidence and the theory evidence disagree [23]. Stopping potential is linear in frequency, $V_{\text{stop}} = (h/e)(\nu - \nu_0)$, with $R^2 = 0.996$ and a slope that recovers h/e to 0.3% (Fig. 4e), so the four-form component accepts the shape. On the same four measurements the theory-conditioned component rejects the classical prediction of frequency-independent photoelectron energy. Their fitted intercept gives $\nu_0 = 4.39 \times 10^{14}$ Hz, and because all four frequencies lie above ν_0 the output is whole-range incompatibility.

Independently measured Solar-System orbits recover Kepler’s exponent of $3/2$ [24] to four figures (1.500) and score 0.02, the four Galilean moons give exponent 1.500 and score 0.00, and Boyle’s original 1662 pressure–volume table [25] gives exponent -0.998 and scores 0.38 (Supplementary Fig. 1a–c). All three classical controls fall below the 0.5 screening threshold.

2.5 Advancing the horizon on the same measurements

We scored each dataset twice on the same grid and under the same error model, first under its old incumbent prediction and then under the historically successful successor, the prediction of the theory that replaced the incumbent; we call this advancing the horizon. The old horizon is rejected in all five cases (bootstrap $p = 0.001$; Fig. 5a). The advanced horizon clears four of the five: FIRAS under Planck’s law [26] ($p = 1.000$), Bertozzi under relativistic kinetic energy [27] ($p = 0.089$), Onnes under a superconducting transition model ($p = 1.000$) and Millikan under the photoelectric relation [28] ($p = 0.900$), with discrepancy statistics falling by factors from 2.0×10^4 to 7.3×10^{10} .

Copper heat capacity, however, is the exception and is sensitive to the error model. A Debye-plus-electronic form [29] reduces the discrepancy by 3.86×10^3 but remains rejected under a 5% reference-curve error floor, the minimum relative error assumed for the reference polynomial ($p = 0.001$), clearing at 7.5%. Across all five cases of this retrospective test, whose alternatives were chosen after the datasets were known (Supplementary Note 11), low-dimensional wrong-shape families remain rejected after Benjamini–Hochberg correction ($q \leq 0.0015$), as do dimensionally coherent but mechanism-irrelevant physical forms (1,000 bootstrap replicates; $p = q = 1/1001$). Four of the five discrepancies therefore collapse under the successful successor prediction and under none of the tested alternatives.

2.6 Measured series at the same horizon

Frozen components applied to sixteen unmodified NIST measurement series give AUROC 0.62 [0.29, 0.89] for the learned score and 0.78 [0.38, 1.00] for the four-form score for the five post-1900 mechanism groups against the nine classical and metrological groups, both intervals including 0.5 (Fig. 6 and Supplementary Note 18). A mechanism group is the set of series produced by one physical mechanism, a metrological group records a calibration artefact or instrument transfer curve, and groups are the units over which the AUROC is computed and resampled. Column semantics, incumbent and mechanism group are declared from each primary source before scoring, both components are used frozen, and all 128 view scores are released (Methods).

Copper thermal expansion measured from 24 K to 852 K takes the highest post-1900 learned score; its coefficient falls towards zero below the Debye temperature, whereas classical equipartition of lattice modes gives a temperature-independent coefficient. The density profile of a cold Bose gas [30, 31] takes the highest four-form score. A sulfur Rydberg series [32], whose inverse-square form was available empirically in 1888, belongs to the classical class yet ranks above every post-1900 group on the learned score (0.1588 against a highest post-1900 group value of 0.1586; Supplementary Note 18).

Circular-interference transmittance rises to a sharp resonance at 451 nm, the most unusual shape in the collection, yet takes a learned score of 0.0630 and a four-form score of 1.946, below the frozen four-form threshold (3.391320708) but above the classical median of 0.566 (Supplementary Note 18). Superconducting magnetization relaxation (flux creep [33]), which no pre-1900 theory accounts for, is close to logarithmic in time and takes the lowest four-form score of any series. Across the whole collection the two highest learned scores belong to a five-year standard-resistor record and a 23-point polymer-impedance curve.

2.7 Calibration, thresholds and the observation process

On a 36-dataset synthetic benchmark on which every method receives the same inputs, the four-form component reaches AUROC 0.80 [0.64, 0.94], against 0.58 for Gaussian-process discrepancy [34] and 0.77 for split-conformal prediction (Supplementary Fig. 1d,f). Withholding the classical law, or both the law and the interior regime, leaves AUROC at 1.00 and 0.97.

We fixed the four-form threshold separately at $n = 5, 9, 43$ and 120 on classical calibration simulations and evaluated it over 12 breakdown and 12 classical families. At $n = 120$, the true-positive rate is 0.271 [0.073, 0.500] at the frozen 5% quantile and 0 at the 1% quantile,

with a realized false-positive rate of 0/96 at both. Sensitivity is 1.0 for smooth roll-off and discontinuous-drop simulations and zero for seven families.

We then altered the observation process while holding the underlying classical relation fixed, to find where that specificity fails. At the strongest simulated levels a detector calibrated to 5% reports false-positive rates of 0% for calibration drift, 40% for sensor saturation, 63% for censoring, 9% for x-axis error, 11% for outliers, 2.5% for correlated noise and 0% for heteroscedasticity (Fig. 5b). The reported output therefore includes a separate field stating whether saturation and censoring were tested, and they are tested explicitly for plateau and threshold cases.

Boundary intervals are calibrated in the same way. Ordinary measurement bootstrap intervals cover only 53.9% of true onsets at a nominal 90% level, and 8.5% for steps. Interval radii calibrated separately at each sample size on 899 independent simulations raise held-out coverage to 91.3% [89.3%, 93.0%] across 900 simulations, with family-wise coverage from 87.8% for smooth turnovers to 94.4% for kinks (Fig. 5c). The reported boundary is therefore an interval.

Simulations extend the evaluation into interiors absent from the FIRAS and Bertozzi measurements. Blackbody radiation spanning $x = h\nu/kT$ from well below to well above unity contains a Rayleigh–Jeans interior with $R^2 = 0.99$ followed by a departure with split-conformal nonconformity 468 and extrapolation-residual ratio 192; relativistic kinetic energy yields a localized boundary at 0.86c, and simulated photoelectric voltage localizes the frequency threshold.

3 Discussion

We introduced a test of a declared knowledge horizon and applied it at 1899. Trained only on source-audited pre-1900 relations, the large encoder ranks 55 twentieth-century law families above nine held-out pre-1900 groups at AUROC 0.925, against 0.649 for the same architecture before optimization. Regenerating those nine controls under the 55-family benchmark’s sampling and noise process leaves 0.857, and the independently frozen four-form component reaches 0.888 against 11 held-out pre-1900 families on the same 55-family benchmark.

A discrepancy is interpretable only when it is tied to a named incumbent. When each historical dataset is scored twice on the same grid and under the same error model, four of five old-horizon discrepancies disappear once the successful successor prediction is admitted, while wrong-shape and dimensionally coherent wrong-mechanism alternatives stay rejected after multiplicity correction. Copper heat capacity is the exception: the advanced model clears only when the reference-curve error floor is relaxed from 5% to 7.5%, so the copper verdict depends on the assumed floor, which therefore has to be declared with the incumbent rather than treated as a nuisance parameter. For a closed-form theory, advancing the horizon amounts to replacing the incumbent prediction and recalibrating its null distribution, and simulator-based references give the corresponding construction where no closed form exists.

Shape novelty and predictive failure are not the same quantity. Later affine, power and exponential laws often stay close to one of the four generic forms (learned and four-form AUROC 0.629/0.795). Threshold and piecewise relations separate on the four-form score (0.982) but not on the learned score (0.600), and nonmonotone and oscillatory relations separate on both (0.909 and 0.864). Conversely, a discrepancy with a named incumbent can keep an ordinary shape, as Millikan’s stopping potential and the Pound–Rebka redshift both do. A classical circular-interference resonance scores 0.0630 on the learned component and stays below the frozen four-form threshold, while superconducting flux creep [33], which no pre-1900 theory accounts for, is close to logarithmic in time and sits at the bottom of the four-form ranking. Keeping the component scores separate preserves this distinction between an unusual shape and the failure of a named prediction.

The negative class of the 1899 experiment is small: nine held-out pre-1900 groups carry the primary learned comparison and set the width of its interval, fourteen mechanism groups set the width of the measured-series intervals, and the generator-matched analysis was registered after the primary scores were opened. The native generator-gated result of 0.536 [0.424, 0.643]

(Results) shows that the trained model is near chance when the later formulas are drawn by the internal generator; the generator-matched value of 0.857 controls for sampling within the curated benchmark only, and the two results are not interchangeable. On measured series the ranking falls to 0.62 for the learned component and 0.78 for the four-form component, with intervals that span chance. The two highest learned scores in that collection belong to a standard-resistor record and a polymer-impedance curve, a pattern consistent with the observation process setting the largest learned-score differences on measured data. Whether this is the sensitivity that saturation and censoring expose for the four-form component in the simulated instrument tests remains to be tested. At the calibrated 5% threshold the four-form component detects breakdown datasets at a true-positive rate of 0.271 [0.073, 0.500] with 0/96 false positives; ranking and calibrated detection are therefore reported as separate results.

EPOCH draws on neural extrapolation [1, 2], split-conformal prediction, Bayesian model discrepancy and change-point detection [3–5], and shares machinery with model-independent searches in particle physics[35] and with machine rediscovery of natural laws [6, 7, 36]. What differs is the target of inference: EPOCH tests the predictive domain of a declared model before asking symbolic regression or theory construction for a replacement. Once a calibrated discrepancy exists, that search can proceed under a parsimony prior [37], as the Bayesian machine scientist and AI Feynman do in recovering Planck-type expressions from blackbody-type data [6, 7], and physical interpretation then supplies the mechanism behind the fitted form.

Large libraries of measured relations could be screened against a frozen horizon, and selected cases rerun with explicit measurement errors and incumbent predictions to obtain a model test and, where an interior is sampled, a boundary interval. Moving the horizon to another period requires a different dated corpus, admitted prediction, observation model and calibration set. The next experiment is to freeze the full decision path in advance and run it against mechanism-independent measurements whose instrument models have been audited.

4 Methods

4.1 Terms

Knowledge horizon: a calendar date; only physical relations recorded before it are admitted to the training corpus and to the laws a test may fit; here 1899. *Incumbent*: the pre-horizon law that a test examines for a given dataset; the incumbent prediction is its numerical prediction for the measured quantity. *Interior* and *frontier*: the contiguous stretch of a dataset over which one admitted pre-horizon form fits well, and the remainder, on which the extrapolation of that fit is tested. *Point cloud*: the (x, y) pairs of one relation, normalized to remove scale and stripped of formula, units, labels and ordering; the only input the learned and four-form components receive. *Law family* and *mechanism group*: the set of point clouds derived from one physical law traced to one dated source, and the set of measured series produced by one physical mechanism; both are the units over which AUROCs are computed and bootstrap intervals resampled. *Four-form component*: fits the best of power, exponential, linear and constant forms on the interior, extrapolates to the frontier and reports the split-conformal nonconformity (the mean absolute frontier residual divided by the 90th percentile of the interior absolute residuals) and the extrapolation-residual ratio (the median absolute log-residual at the frontier divided by its interior value), both with a one-percent relative floor in the denominator; called the data-anchored or statistical component in the code and SI. *Theory-conditioned component*: computes the residuals of the measurements from the incumbent prediction, scaled by the declared measurement errors, and compares their size with a null distribution simulated from that prediction on the same grid with the same errors. *Learned component*: a neural encoder trained only on point clouds from pre-horizon relations, with no example of a failure (positive-only); its score is the distance of a new cloud’s embedding from those of the training relations; the large encoder has 20.9 million parameters and the compact encoder has width 128 and uses no unit information. *Null distribution*: the distribution of a discrepancy statistic over many datasets simulated from the incumbent prediction on the same x -grid and with the same error model as the real measurements. *Observation model*: the declared description of how the measurements were

produced, comprising error distribution, censoring or saturation, and grid; an error floor is a minimum relative error assigned when no published covariance exists. *Frozen*: fixed, and recorded with a time stamp and file hash, before any test data were scored. *Routing rule*: the fixed rule that decides before scoring which component evaluates a dataset. *Verdicts*: compatible (the incumbent fits over the measured range); localized incompatibility (the incumbent fits an interior and fails beyond a boundary reported with an interval); whole-range incompatibility (the incumbent fails everywhere sampled); abstention (the inputs do not support a test). *Advancing the horizon*: rescored the same measurements on the same grid and with the same error model after replacing the incumbent prediction by the prediction of the theory that historically superseded it. *Generator-matched* and *native controls*: the nine pre-1900 control groups regenerated with the sampling and noise process of the later families, and the same groups as originally generated; the generator-gated set draws later formulas from the internal archive by the same generator as the pre-1900 corpus.

4.2 Declaring a test

An EPOCH test specifies $\mathcal{T} = (\mathcal{M}, \Theta, P_\varepsilon, I, A, S, \alpha)$. Here $\mathcal{M}$ lists the admissible models, Θ the fitted parameters, P_ε specifies the observation and censoring errors, I identifies the fitting data, A specifies every adaptive choice, S is the discrepancy statistic and α is the error rate. Because direction choice, window search, model selection and parameter fitting all belong to A , each of them is rerun inside every calibration and null replicate rather than being fixed once on the test data. Confirmatory tests use the finite-sample upper-tail probability $p = (1 + \#\{S_i \geq S_*\})/(n_{\text{cal}} + 1)$ with ties counted conservatively, and dataset scans report Benjamini–Hochberg adjusted q -values [38]. The primary endpoint is the true-positive rate at a threshold targeted to 5% false positives on independent calibration, with the 1% threshold secondary; the 0.5 percentile used in the historical screen is a screening threshold. The finite-sample guarantee of the upper-tail test holds when calibration and test units are exchangeable; that unit is the law family (the source-law family of the SI) or the mechanism group, not the individual point or frontier.

4.3 The four forms and the interior and frontier

The four-form component (called the data-anchored or statistical component in the code and SI) draws on a fixed vocabulary of power, exponential, linear and constant forms. An eight-form ablation adding quadratic, cubic, square-root and logarithmic terms was tested and discarded, because the four-form version extrapolates better on validation data. Given a point cloud, a fixed rule takes the candidate interior to be the contiguous regime in which one admitted form fits with high internal R^2 , and the frontier to be the complementary region. The search runs from both ends and fit quality decides whether the low- or the high-value direction is retained. The split-conformal nonconformity [39] is the mean absolute frontier residual divided by the 90th percentile of the interior absolute residuals, and the extrapolation-residual ratio compares the median log-residual at the frontier with its interior value. Each form is fitted from several initializations, and both denominators carry a one-percent relative floor so that a near-perfect interior fit cannot produce an unbounded ratio. A capped variant, which averages rescaled components clipped at 3, is retained for comparison with earlier results. Relations with fewer than 20 points are calibrated against size-matched simulations.

4.4 Large-scale pre-1900 representation

The scaling experiment uses a 20,908,547-parameter encoder trained alongside a second, training-only network, without any anomaly supervision. Its query branch is a permutation-invariant six-block Set Transformer[40] of width 384 with eight heads, four pooling tokens and a 256-dimensional normalized output. During positive-only pretraining it is aligned to that expression-tree network, built from four edge-aware graph updates and two graph-local attention layers, one of whose update networks is a 96-channel eight-basis radial-basis-function Kolmogorov–Arnold network module [41]. At query time the encoder receives an unordered (x, y) cloud normalized by median and interquartile range, and nothing else: no equation, graph, units, date, law name, post-cutoff label or deformation example.

Training draws on 880,896 clouds derived from 3,441 audited formula reductions of pre-1900 laws, each in dimensionless one-dimensional form. These are grouped into 91 sets of clouds treated as the same law during training, spanning 64 historical-family tags. Held-out calibration and internal sets contain 11,648 and 62,336 clouds from 91 and 487 formulas. Reducing these to scientific groups leaves nine historical-family groups, which form the internal control class for every AUROC reported here. The model trains for 30 epochs at batch size 1,024 under a checkpoint rule fixed before any temporal evaluation. Its learned score (a one-class score in the vocabulary of anomaly detection) averages four calibrated ranks, each transformed by its own pre-1900 calibration empirical cumulative distribution function: top-eight cosine distance to graph-formula embeddings, top-eight distance to identity centroids, top-eight distance to point-formula prototypes, and shrinkage-Mahalanobis distance [42, 43] to those prototypes. Clouds are reduced by median to formulas and then to law families or frozen relation groups, and family- and group-level AUROCs use 2,000 grouped-bootstrap replicates. The same network before optimization passes through the identical pipeline. Code, configuration, score construction, seed and checkpoint rule were frozen before the first score on later physics was computed.

For the generator-matched analysis, all nine internal control families were regenerated with the 55-family benchmark’s point count, x-jitter, repeat count, noise range and heteroscedastic profile, while the checkpoint, reference memories, calibration distribution function, score and aggregation were left unchanged. The same analysis was then repeated with the checkpoint before optimization. A separate 96,064-cloud generator-gated set, in which later formulas are drawn from the internal archive by the same generator as the pre-1900 corpus instead of from the citation-anchored registry, and the cumulative pre-1950 pilot are reported in Supplementary Note 16.

4.5 Compact encoder and its controls

The compact encoder is a set transformer of width 128 with three induced-set-attention blocks, and trains for 1,000 steps on two independently subsampled and rescaled views of 205 source-audited classical point clouds drawn from 40 families. A record is one generated dataset of a law; a view is one resampled or rescaled copy of a record or of a measured series. Families are sampled without replacement so that repeated laws cannot become false contrastive negatives, and an auxiliary head predicts functional signatures. No breakdown, saturation, step, kink, splice or crossover example enters this optimization. Matched models are trained with SI dimensions and with zeroed dimension metadata, and the zero-unit model is primary for the synthetic benchmark. Frozen k-nearest-neighbour and shrinkage-Mahalanobis learned scores are compared against a matched random encoder. Leave-family-out calibration holds out each of the 40 training families in full, reduces the eight views within a record and then reduces records to one median per family. The threshold, the 95th percentile of the 40 held-out family scores, is frozen before evaluation, and the training and evaluation files carry each other’s hashes, which are checked before scoring.

The generic-function control encoder is a second zero-unit encoder that shares architecture, seed, batch size, augmentation, auxiliary objective and 1,000-step schedule but is trained on 205 point clouds from globally valid smooth expressions in 40 pseudo-families, matched to the historical set in record count, family-size profile and the histogram of eight-bit functional signatures. Both encoders are retrained from scratch under three paired initialization and augmentation seeds while the evaluation-view seed and the 24 breakdown and classical families stay fixed, and we report the observed range and sample standard deviation across those runs without altering the canonical checkpoints. Finally, each of 46 held-out point clouds is paired with saturation, step, kink, splice, roll-off and crossover deformations that retain its archived input and output SI vectors. Scores reduce to one clean and six deformation values within each of seven law families, the AUROC difference is bootstrapped by family, and each checkpoint is read against its own threshold, the 95th percentile of its held-out family scores. Thirty-four of those records carry at least one nonzero dimensional exponent.

Routing for the learned component is fixed in advance. Point clouds with at least 20 observations receive a learned score, point clouds whose response is nearly constant are sent instead to the theory-conditioned component, which tests a constant against noise, and formula and deformation families are the resampling units throughout.

4.6 Frozen post-1900 formula transfer

The transfer registry (the 55-family benchmark) contains 55 law families first valid between 1901 and 1996, each carrying a primary historical citation and 45 of them a DOI. The families include thermionic emission [44], discrete atomic excitation [45], parity violation [46], absence of diffusion in disordered lattices [47], the local-hidden-variable bound and its experimental test [48, 49] and the first gravitational-wave detection [50]. Every law is represented by a disclosed one-dimensional form $y = f(x)$ in dimensionless variables over a fixed support, realized eight times at 120 points with varying nuisance parameters, sampling and 0.5–2.5% noise, giving 440 records across 55 families. Special reductions, such as the BCS interpolation [51], the two-flavour neutrino survival curve [52] and the ideal quantum-Hall staircase [53], are marked as such in the registry. Offline checks cover family and seed uniqueness, citation-year consistency, finite nonconstant curves, point ordering, file hashes and separation from the pre-1900 archive.

Transfer uses the previously fixed model and calibration. The checkpoint and the pre-1900 data hash must match those of the cross-family evaluation before scoring begins. The learned score is reduced by median over views, records and families and read against the pre-existing threshold from the 40 held-out family scores. The four-form score reruns the full four-form selection and extrapolation procedure against the existing $n = 120$ thresholds. Eleven cited pre-1900 families are the controls, AUROC intervals resample families, and twelve of the post-1900 laws deliberately overlap the declared affine, power or exponential forms. Per-family conformal p -values and component union and intersection rates are reported as secondary diagnostics.

4.7 Historical measurements and horizon-specificity controls

We analyse the published COBE-FIRAS monopole spectrum with its pointwise errors, five Bertozzi values digitized from the 1964 report, NIST specific-heat reference polynomials for six materials, Onnes resistance values including upper limits, Millikan sodium stopping potentials and Solar-System orbital constants. Each case is stored as a machine-readable record of the test, recording input status, model set, fitted parameters, observation model, fit subset, adaptive procedure, decision rule, output and provenance. Old- and advanced-horizon null distributions are evaluated on the same x -grid, and explicit error floors stand in for unavailable covariance and for digitization uncertainty and are varied in sensitivity analysis.

To establish that a collapse in the discrepancy is specific to the successful successor, wrong-shape and dimensionally coherent wrong-mechanism forms are refitted on the same case grids and observation models with 1,000 parametric-bootstrap replicates. The physical controls substitute fermionic occupation for photon statistics, a proton for the electron rest-energy scale, an Einstein oscillator [54] for Debye acoustic modes [29], Arrhenius transport for the superconducting transition, and Compton recoil for the photoelectric slope; dimensional ratios and parameter units are retained in the machine-readable output.

4.8 Measured series

Sixteen series are taken unmodified from the NIST Dataplot and Statistical Reference Dataset collections and bound to their SHA-256 hashes. A registry declares, for each series, the numeric block, the abscissa and response columns, the 1899 incumbent, the mechanism group and whether the mechanism was known before 1899, all before any score is computed. Where a file header and its stored block disagree on column order, the order is resolved against certified values, and a file that is a subset or decimation of another series shares that series' mechanism group. Series are sorted by abscissa and thinned uniformly to at most 2,000 points. The learned component applies the fixed rule that sends nearly constant responses to the theory-conditioned component, rescales each measured series and adds no synthetic noise, with the reference bank rebuilt from the frozen pre-1900 training split. The four-form component reruns the complete four-form direction, window, model-selection and scoring procedure on eight views drawn at the largest frozen calibration size each series supports. Scores reduce by median within a series and then within a mechanism group, and the grouped AUROC resamples mechanism groups.

540 4.9 Synthetic suites, baselines and localization

541 The held-out evaluation comprises 180 labelled datasets and the fair baseline suite 36, on which
542 split-conformal prediction, the extrapolation-residual ratio, Gaussian-process discrepancy [34],
543 CUSUM [4] and RESET [55] are compared. The mechanism-level benchmark contains 12
544 classical and 12 breakdown families with eight records each; its 5% and 1% thresholds
545 come from 785 valid size-matched calibration simulations, and uncertainty is resampled by
546 breakdown or classical family. The learned checkpoint evaluation uses 192 datasets over the
547 same 24 families. Instrument tests apply calibration drift, sensor saturation, censoring, x-axis
548 error, outliers, correlated noise and heteroscedasticity to classical synthetic relations while
549 leaving the underlying law intact, and boundary intervals are obtained by split-conformal
550 calibration of the normalized localization error separately at $n = 40, 80$ and 160.

551 4.10 Real-measurement inventory

552 Forty-four files were downloaded from the NIST Dataplot and Statistical Reference Dataset
553 collections and their SHA-256 hashes frozen. The inventory holds 20 instrument controls,
554 11 classical controls, six controls whose post-1900 mechanism is known and seven horizon
555 cases. A dataset is promoted into the evaluation registry only once its column semantics,
556 incumbent prediction, measurement model, provenance and mechanism-level grouping have
557 been audited.

558 5 Data availability

559 The public COBE-FIRAS, Solar-System, Bertozzi, NIST, Millikan and Kamerlingh Onnes in-
560 puts are identified in the Supplementary Information. The data package contains source URLs
561 and hashes; the 55-family formula registry and 440 generated point clouds; the 11-family de-
562 velopment registry and 88 generated records; the pre-1900 training, calibration and evaluation
563 manifests; and every numerical result used in the figures. The project repository is avail-
564 able at <https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons>,
565 and the archive for this release at <https://doi.org/10.5281/zenodo.22742368>; <https://doi.org/10.5281/zenodo.22138620> resolves to the most recent version.

567 6 Code availability

568 Code is available at [https://github.com/ChenxiHeCam/](https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons)
569 [Epistemic-probe-of-Classical-Horizons](https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons). The repository contains corpus audits,
570 point-cloud generators, frozen evaluators, statistical calibration and robustness experiments,
571 model definitions, training configurations, figure scripts and the evidence-manifest validator.
572 Checkpoint and data hashes are verified before evaluation, and the release archived
573 at <https://doi.org/10.5281/zenodo.22742368> is the version used for every number
574 reported here.

575 7 Acknowledgements

576 We thank NASA LAMBDA, the NIST Cryogenic Material Properties database and the
577 KNAW digital archive for public data access. This research received no specific grant from
578 any funding agency in the public, commercial, or not-for-profit sectors.

579 8 Author contributions

580 C.H. conceived the framework, designed and built EPOCH, ran the experiments, and wrote
581 the manuscript. J.C. contributed to the benchmark construction, case-study analysis, and
582 manuscript revision. Both authors reviewed and approved the final manuscript.

9 Competing interests 583

The authors declare no competing interests. 584

Correspondence and requests for materials should be addressed to C.H. 585
(ch2067@cam.ac.uk). 586

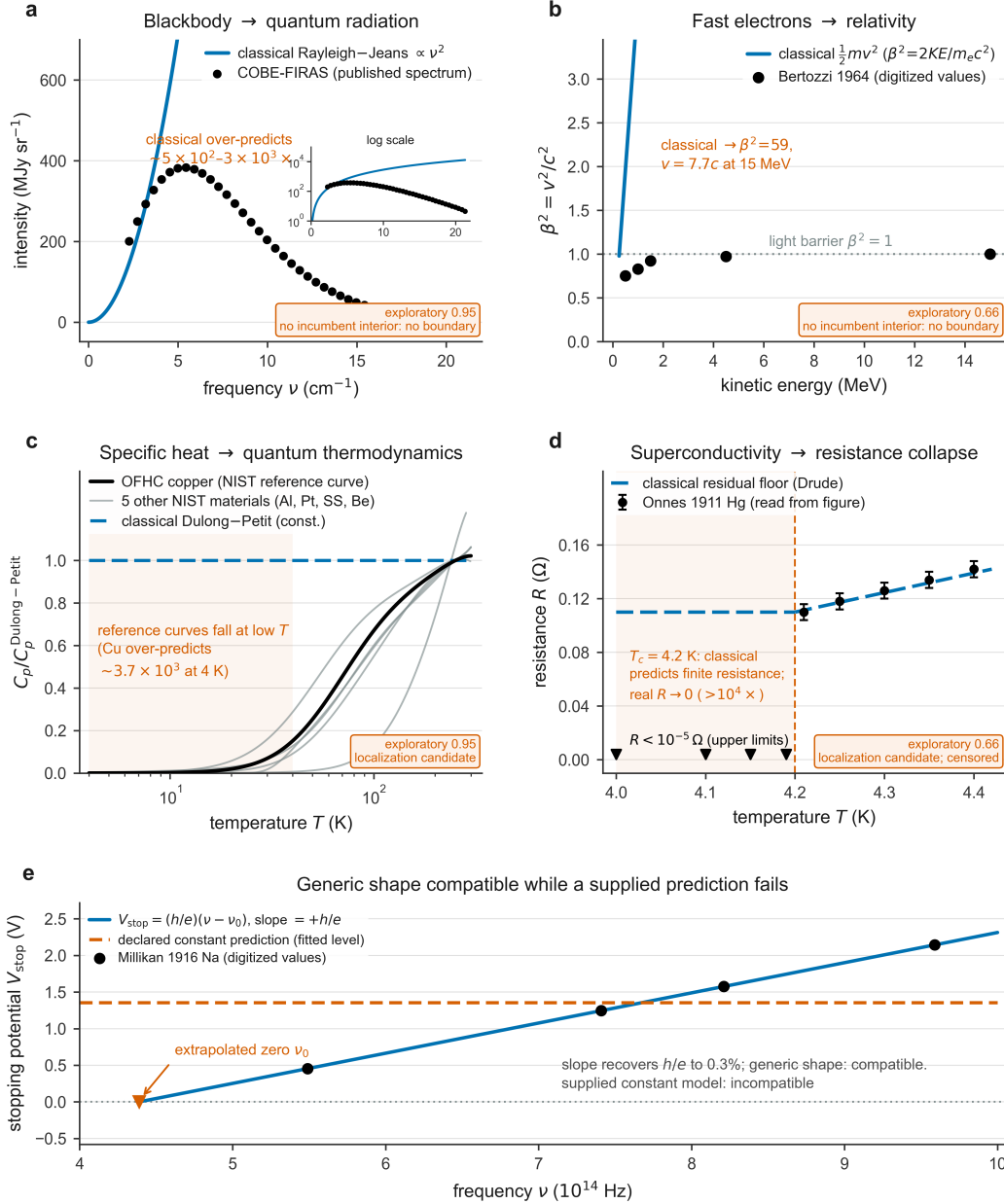


Figure 4. Historical measurements under the 1899 horizon. a–d, Black points show published processed measurements, digitized historical values or evaluated reference curves as labelled; blue curves show incumbent predictions. Orange badges give screening percentiles. FIRAS and Bertozzi yield whole-range incompatibility. Specific heat and Onnes contain an interior, allowing a candidate boundary to be marked by the shaded frontier. Tail probabilities and multiplicity calculations are reported in Supplementary Note 4. e, In Millikan’s 1916 digitized values, stopping potential rises linearly with frequency as $V_{stop} = (h/e)(\nu - \nu_0)$. The slope recovers h/e to 0.3%, and the marker denotes the extrapolated zero crossing ν_0 . The four-form score falls below the threshold, whereas the theory-conditioned component rejects the classical prediction of frequency-independent energy (dashed line).

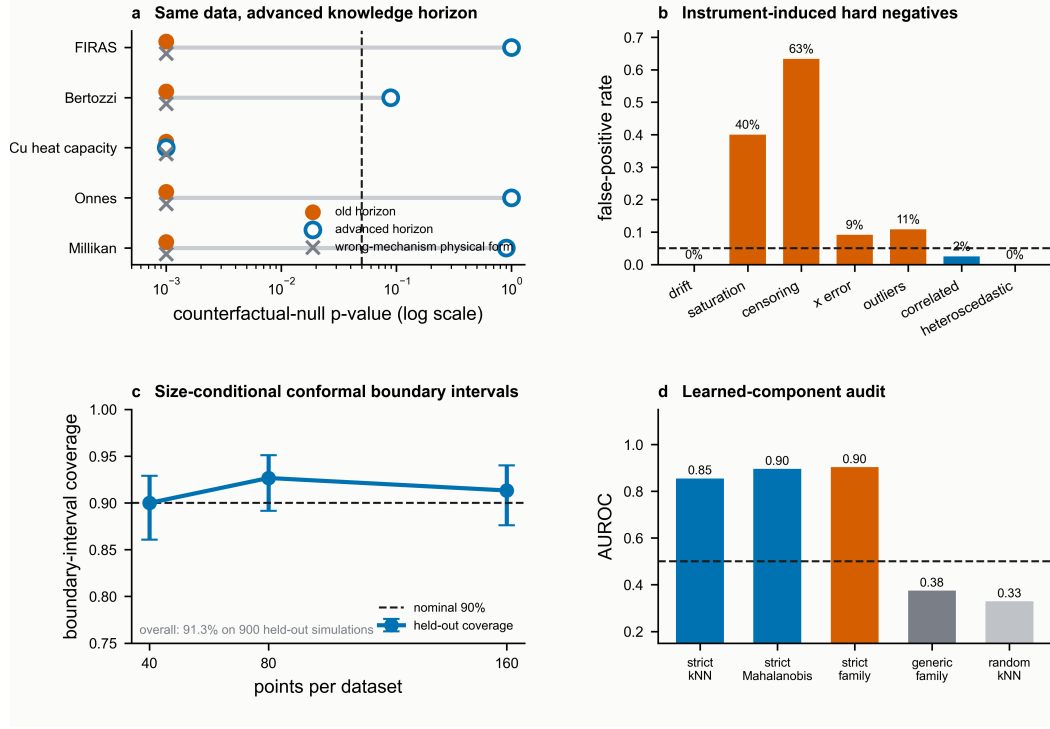


Figure 5. Calibration and controls for the EPOCH components. **a**, Same-grid goodness-of-fit p -values under old and advanced horizons. Four of five advanced predictions clear at 5%; five wrong-mechanism forms remain rejected. **b**, False-positive rates under simulated instrument effects, with the largest changes caused by saturation and censoring. **c**, Held-out coverage of 90% boundary intervals calibrated separately at each sample size. **d**, Compact encoder trained only on 205 records from 40 pre-1900 families with no breakdown example. Family-grouped AUROC is 0.903 [0.729, 1.000] against 0.375 [0.167, 0.618] for the generic-function control encoder on the same evaluation families; dataset-level AUROC is 0.854 by k -nearest-neighbour distance and 0.895 by shrinkage Mahalanobis distance, against 0.329 and 0.288 for matched random features (Results).

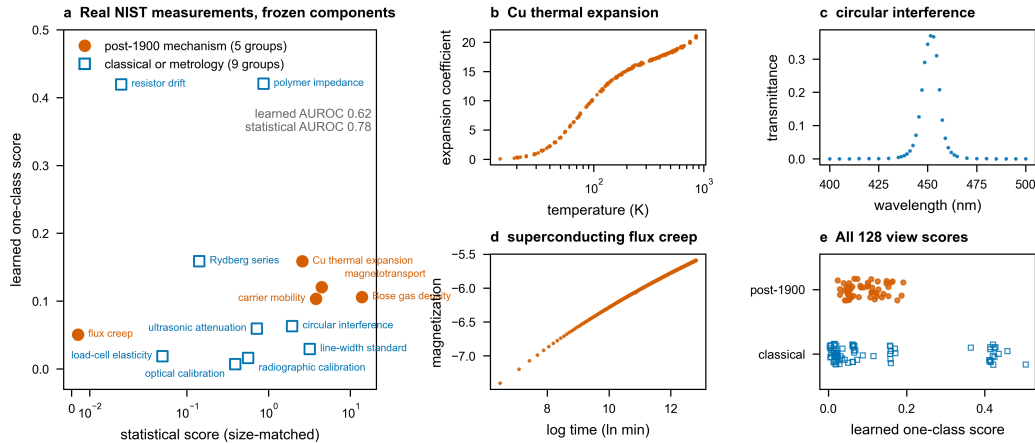


Figure 6. Frozen components on measured series. **a**, Mechanism-group medians of the learned and four-form scores for sixteen NIST measurement series in fourteen groups, five outside the pre-1900 model set and nine classical or metrological. **b–d**, Three of the measured series: copper thermal expansion from 24 K to 852 K, circular-interference transmittance, and superconducting magnetization relaxation against log time. **e**, Every individual measurement-view score behind the medians in **a**. The released quantity is the continuous component output.

References

- [1] Hanchen Wang, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, Payal Chandak, Shengchao Liu, Peter Van Katwyk, Andreea Deac, Anima Anandkumar, Karianne Bergen, Carla P. Gomes, Shirley Ho, Pushmeet Kohli, Joan Lasenby, Jure Leskovec, Tie-Yan Liu, Arjun Manrai, Debora Marks, Bharath Ramsundar, Le Song, Jimeng Sun, Jian Tang, Petar Veličković, Max Welling, Linfeng Zhang, Connor W. Coley, Yoshua Bengio, and Marinka Zitnik. Scientific discovery in the age of artificial intelligence. *Nature*, 620:47–60, 2023.
- [2] Keyulu Xu, Mozhi Zhang, Jingling Li, Simon S. Du, Ken-ichi Kawarabayashi, and Stefanie Jegelka. How neural networks extrapolate: from feedforward to graph neural networks. In *International Conference on Learning Representations*, 2021.
- [3] Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning*, 16:494–591, 2023.
- [4] E. S. Page. Continuous inspection schemes. *Biometrika*, 41:100–115, 1954.
- [5] Marc C. Kennedy and Anthony O’Hagan. Bayesian calibration of computer models. *Journal of the Royal Statistical Society B*, 63:425–464, 2001.
- [6] Roger Guimerà, Ignasi Reichenardt, Antoni Aguilar-Mogas, Francesco A. Massucci, Manuel Miranda, Jordi Pallarès, and Marta Sales-Pardo. A Bayesian machine scientist to aid in the solution of challenging scientific problems. *Science Advances*, 6:eaav6971, 2020.
- [7] Silviu-Marian Udrescu and Max Tegmark. AI Feynman: a physics-inspired method for symbolic regression. *Science Advances*, 6:eaay2631, 2020.
- [8] Ilya Frank and Igor Tamm. Coherent visible radiation of fast electrons passing through matter. *Comptes Rendus de l’Académie des Sciences de l’URSS*, 14:109–114, 1937.
- [9] Ugo Fano. Effects of configuration interaction on intensities and phase shifts. *Physical Review*, 124:1866–1878, 1961. doi: 10.1103/PhysRev.124.1866.
- [10] Adam G. Riess, Alexei V. Filippenko, Peter Challis, Alejandro Clocchiatti, Alan Diercks, Peter M. Garnavich, Ron L. Gilliland, Craig J. Hogan, Saurabh Jha, and Robert P. Kirshner. Observational evidence from supernovae for an accelerating universe and a cosmological constant. *The Astronomical Journal*, 116:1009–1038, 1998. doi: 10.1086/300499.
- [11] Robert V. Pound and Glen A. Rebka. Apparent weight of photons. *Physical Review Letters*, 4:337–341, 1960. doi: 10.1103/PhysRevLett.4.337.
- [12] Hendrik B. G. Casimir. On the attraction between two perfectly conducting plates. *Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen*, 51:793–795, 1948.
- [13] Saul Perlmutter, Greg Aldering, Gerson Goldhaber, R. A. Knop, Peter Nugent, P. G. Castro, S. Deustua, S. Fabbro, A. Goobar, and D. E. Groom. Measurements of ω and λ from 42 high-redshift supernovae. *The Astrophysical Journal*, 517:565–586, 1999. doi: 10.1086/307221.
- [14] Willis E. Lamb and Robert C. Retherford. Fine structure of the hydrogen atom by a microwave method. *Physical Review*, 72:241–243, 1947. doi: 10.1103/PhysRev.72.241.
- [15] Julian Schwinger. On quantum-electrodynamics and the magnetic moment of the electron. *Physical Review*, 73:416–417, 1948. doi: 10.1103/PhysRev.73.416.
- [16] D. J. Fixsen, E. S. Cheng, J. M. Gales, J. C. Mather, R. A. Shafer, and E. L. Wright. The cosmic microwave background spectrum from the full COBE FIRAS data set. *The Astrophysical Journal*, 473:576–587, 1996.
- [17] Lord Rayleigh. Remarks upon the law of complete radiation. *Philosophical Magazine*, 49:539–540, 1900.

- [18] James H. Jeans. On the partition of energy between matter and aether. *Philosophical Magazine*, 10:91–98, 1905. 635 636
- [19] William Bertozzi. Speed and kinetic energy of relativistic electrons. *American Journal of Physics*, 32:551–555, 1964. 637 638
- [20] Pierre Louis Dulong and Alexis Thérèse Petit. Recherches sur quelques points importants de la théorie de la chaleur. *Annales de Chimie et de Physique*, 10:395–413, 1819. 639 640
- [21] Heike Kamerlingh Onnes. Further experiments with liquid helium. G. On the electrical resistance of pure metals, etc. VI. On the sudden change in the rate at which the resistance of mercury disappears. *Communications from the Physical Laboratory of the University of Leiden*, 124c, 1911. 641 642 643 644
- [22] Paul Drude. Zur elektronentheorie der metalle. *Annalen der Physik*, 306:566–613, 1900. doi: 10.1002/andp.19003060312. 645 646
- [23] R. A. Millikan. A direct photoelectric determination of Planck’s “h”. *Physical Review*, 7:355–388, 1916. 647 648
- [24] Johannes Kepler. *Harmonices Mundi, Book V*. 1619. 649
- [25] Robert Boyle. *A Defence of the Doctrine Touching the Spring and Weight of the Air*. Thomas Robinson, Oxford, 1662. 650 651
- [26] Max Planck. Über das gesetz der energieverteilung im normalspectrum. *Annalen der Physik*, 309:553–563, 1901. doi: 10.1002/andp.19013090310. 652 653
- [27] Albert Einstein. Zur elektrodynamik bewegter körper. *Annalen der Physik*, 322:891–921, 1905. doi: 10.1002/andp.19053221004. 654 655
- [28] Albert Einstein. Über einen die erzeugung und verwandlung des lichtes betreffenden heuristischen gesichtspunkt. *Annalen der Physik*, 322:132–148, 1905. doi: 10.1002/andp.19053220607. 656 657 658
- [29] Peter Debye. Zur theorie der spezifischen wärmen. *Annalen der Physik*, 344:789–839, 1912. doi: 10.1002/andp.19123441404. 659 660
- [30] Satyendra Nath Bose. Plancks gesetz und lichtquantenhypothese. *Zeitschrift für Physik*, 26:178–181, 1924. doi: 10.1007/BF01327326. 661 662
- [31] Mike H. Anderson, Jason R. Ensher, Michael R. Matthews, Carl E. Wieman, and Eric A. Cornell. Observation of bose–einstein condensation in a dilute atomic vapor. *Science*, 269:198–201, 1995. doi: 10.1126/science.269.5221.198. 663 664 665
- [32] Johannes R. Rydberg. On the structure of the line-spectra of the chemical elements. *Philosophical Magazine*, 29:331–337, 1890. 666 667
- [33] P. W. Anderson and Y. B. Kim. Hard superconductivity: theory of the motion of abrikosov flux lines. *Reviews of Modern Physics*, 36:39–43, 1964. doi: 10.1103/RevModPhys.36.39. 668 669 670
- [34] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006. 671 672
- [35] Raffaele Tito D’Agnolo and Andrea Wulzer. Learning new physics from a machine. *Physical Review D*, 99:015014, 2019. 673 674
- [36] Michael Schmidt and Hod Lipson. Distilling free-form natural laws from experimental data. *Science*, 324:81–85, 2009. 675 676
- [37] Chenxi He and Jingqiu Chen. A search-free foundation model for symbolic regression of physical laws. *Research Square*, 2026. doi: 10.21203/rs.3.rs-10495716/v1. Preprint, not peer reviewed. 677 678 679

- [38] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57:289–300, 1995.
- [39] Vladimir Vovk, Alexander Gammernan, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- [40] Juho Lee, Yoonho Lee, Jungtaek Kim, Adam R. Kosiorek, Seungjin Choi, and Yee Whye Teh. Set Transformer: a framework for attention-based permutation-invariant neural networks. *Proceedings of Machine Learning Research*, 97:3744–3753, 2019.
- [41] Ziming Liu, Yixuan Wang, Sachin Vaidya, Fabian Ruehle, James Halverson, Marin Soljačić, Thomas Y. Hou, and Max Tegmark. KAN: Kolmogorov–Arnold networks. In *International Conference on Learning Representations*, 2025.
- [42] Prasanta Chandra Mahalanobis. On the generalised distance in statistics. *Proceedings of the National Institute of Sciences of India*, 2:49–55, 1936.
- [43] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88:365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.
- [44] Owen W. Richardson. On the negative radiation from hot platinum. *Proceedings of the Cambridge Philosophical Society*, 11:286–295, 1901.
- [45] James Franck and Gustav Hertz. Über zusammenstöße zwischen elektronen und den molekülen des quecksilberdampfes und die ionisierungsspannung desselben. *Verhandlungen der Deutschen Physikalischen Gesellschaft*, 16:457–467, 1914.
- [46] Chien-Shiung Wu, Ernest Ambler, Raymond W. Hayward, Dale D. Hoppes, and Ralph P. Hudson. Experimental test of parity conservation in beta decay. *Physical Review*, 105:1413–1415, 1957. doi: 10.1103/PhysRev.105.1413.
- [47] Philip W. Anderson. Absence of diffusion in certain random lattices. *Physical Review*, 109:1492–1505, 1958. doi: 10.1103/PhysRev.109.1492.
- [48] John S. Bell. On the einstein podolsky rosen paradox. *Physics Physique Fizika*, 1:195–200, 1964. doi: 10.1103/PhysicsPhysiqueFizika.1.195.
- [49] Alain Aspect, Jean Dalibard, and Gérard Roger. Experimental test of bell’s inequalities using time-varying analyzers. *Physical Review Letters*, 49:1804–1807, 1982. doi: 10.1103/PhysRevLett.49.1804.
- [50] Benjamin P. Abbott et al. Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116:061102, 2016. doi: 10.1103/PhysRevLett.116.061102.
- [51] John Bardeen, Leon N. Cooper, and J. Robert Schrieffer. Theory of superconductivity. *Physical Review*, 108:1175–1204, 1957. doi: 10.1103/PhysRev.108.1175.
- [52] Ziro Maki, Masami Nakagawa, and Shoichi Sakata. Remarks on the unified model of elementary particles. *Progress of Theoretical Physics*, 28:870–880, 1962. doi: 10.1143/PTP.28.870.
- [53] Klaus von Klitzing, Gerhard Dorda, and Michael Pepper. New method for high-accuracy determination of the fine-structure constant based on quantized hall resistance. *Physical Review Letters*, 45:494–497, 1980. doi: 10.1103/PhysRevLett.45.494.
- [54] Albert Einstein. Die plancksche theorie der strahlung und die theorie der spezifischen wärme. *Annalen der Physik*, 327:180–190, 1907. doi: 10.1002/andp.19063270110.
- [55] J. B. Ramsey. Tests for specification errors in classical linear least-squares regression analysis. *Journal of the Royal Statistical Society B*, 31:350–371, 1969.